

Executive Sales Performance Dashboard using Python, MySQL & Power BI

End-to-End Data Analytics Portfolio Project

Prepared by:

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Tools:

Python | MySQL | Power BI | Pandas | NumPy

GitHub:

https://github.com/Nehas9975/Executive_Sales_Performance_Dashboard

1. Introduction

This portfolio project demonstrates the complete data analytics workflow from data cleaning and SQL analysis to interactive dashboard development.

The **Executive Sales Performance Dashboard** is an end-to-end Data Analytics project developed using **Python, MySQL, and Power BI**. The objective of this project is to analyze retail sales data from the Superstore dataset and generate meaningful business insights. The project demonstrates the complete analytics workflow, including data cleaning, SQL-based analysis, and dashboard visualization.

Python was used to clean and preprocess the dataset, MySQL was used to validate and analyze business data using SQL queries, and Power BI was used to build an interactive dashboard that presents key performance indicators (KPIs) and business trends.

2. Problem Statement

Retail businesses generate large volumes of sales data every day. Analyzing this data manually is time-consuming and often leads to inefficient decision-making. Businesses require an efficient solution that can clean data, perform business analysis, and visualize important metrics.

This project addresses these challenges by using **Python** for data preprocessing, **MySQL** for SQL-based business analysis, and **Power BI** for interactive dashboard visualization.

3. Objectives

- Clean and preprocess the Superstore Sales dataset using Python.
- Validate and analyze sales data using MySQL.
- Create an interactive dashboard using Power BI.
- Analyze sales across categories, regions, states, cities, and products.
- Generate meaningful business insights.
- Support data-driven business decision-making.

4. Dataset Description

The Superstore Sales dataset contains retail sales transaction records. It includes information about customer orders, product categories, sales, order dates, shipping details, and geographical locations.

The dataset was cleaned using Python, validated through SQL queries in MySQL, and visualized using Power BI.

```
[6]: df.shape
```

```
[6]: (9800, 18)
```

Figure 4.1: Dataset Shape

Dataset Information

- Dataset Name: Superstore Sales Dataset
- Total Records: **9,800**
- Total Columns: **22**
- Total Orders: **4922**
- Total Customers: **793**
- File Format: **CSV**
- Data Source: Kaggle – Superstore Sales Dataset

```
•[8]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9800 entries, 0 to 9799
Data columns (total 18 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Row ID          9800 non-null  int64
1   Order ID        9800 non-null  object
2   Order Date      9800 non-null  object
3   Ship Date       9800 non-null  object
4   Ship Mode       9800 non-null  object
5   Customer ID     9800 non-null  object
6   Customer Name   9800 non-null  object
7   Segment        9800 non-null  object
8   Country         9800 non-null  object
9   City            9800 non-null  object
10  State           9800 non-null  object
11  Postal Code     9789 non-null  float64
12  Region          9800 non-null  object
```

Figure 4.2 Dataset Information

Columns Used

- Order ID
- Order Date
- Ship Date
- Customer ID
- Customer Name
- Category
- Product Name
- Sales

- Region
- State
- City
- Year
- Month
- Quarter

```
[10]: df.isnull().sum()
[10]: Row ID      0
      Order ID   0
      Order Date  0
      Ship Date   0
      Ship Mode   0
      Customer ID  0
      Customer Name 0
      Segment     0
      Country     0
      City        0
      State       0
      Postal Code 11
      Region      0
      Product ID  0
      Category    0
      Sub-Category 0
      Product Name 0
      Sales       0
      dtype: int64
```

Figure 4.3 Missing Value Analysis

5. Tools & Technologies

Tool	Purpose
Python	Data Cleaning & Preprocessing
Pandas	Data Manipulation
NumPy	Numerical Computations
Matplotlib	Data Visualization
Seaborn	Advanced Statistical Data Visualization
MySQL	SQL Analysis & Data Validation
Jupyter Notebook	Python Development
Power BI Desktop	Dashboard Development
CSV	Dataset Storage

Python was used to clean and preprocess the dataset. Pandas and NumPy were used for data manipulation, while Matplotlib was used for exploratory visualization. The cleaned dataset was then imported into MySQL for SQL analysis and validation. Finally, Power BI was used to create an interactive Executive Sales Dashboard.

6. Project Workflow

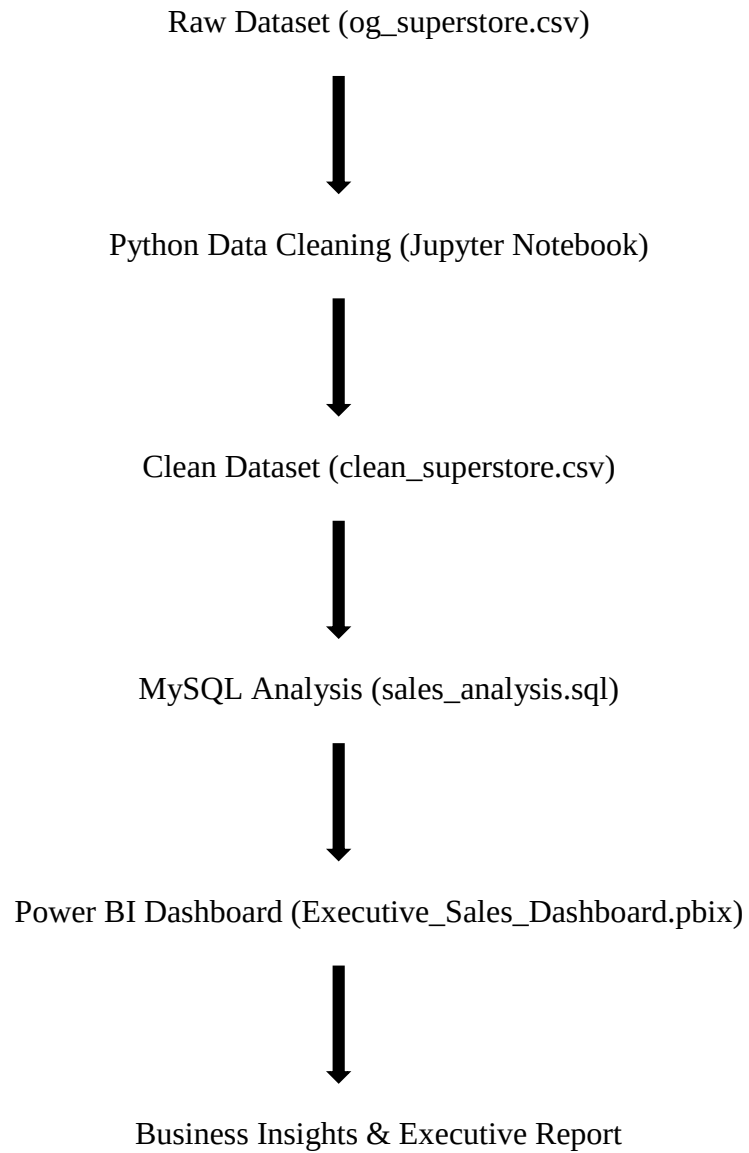


Figure 6.1:End-to-End Data Analytics Workflow

7. Dashboard

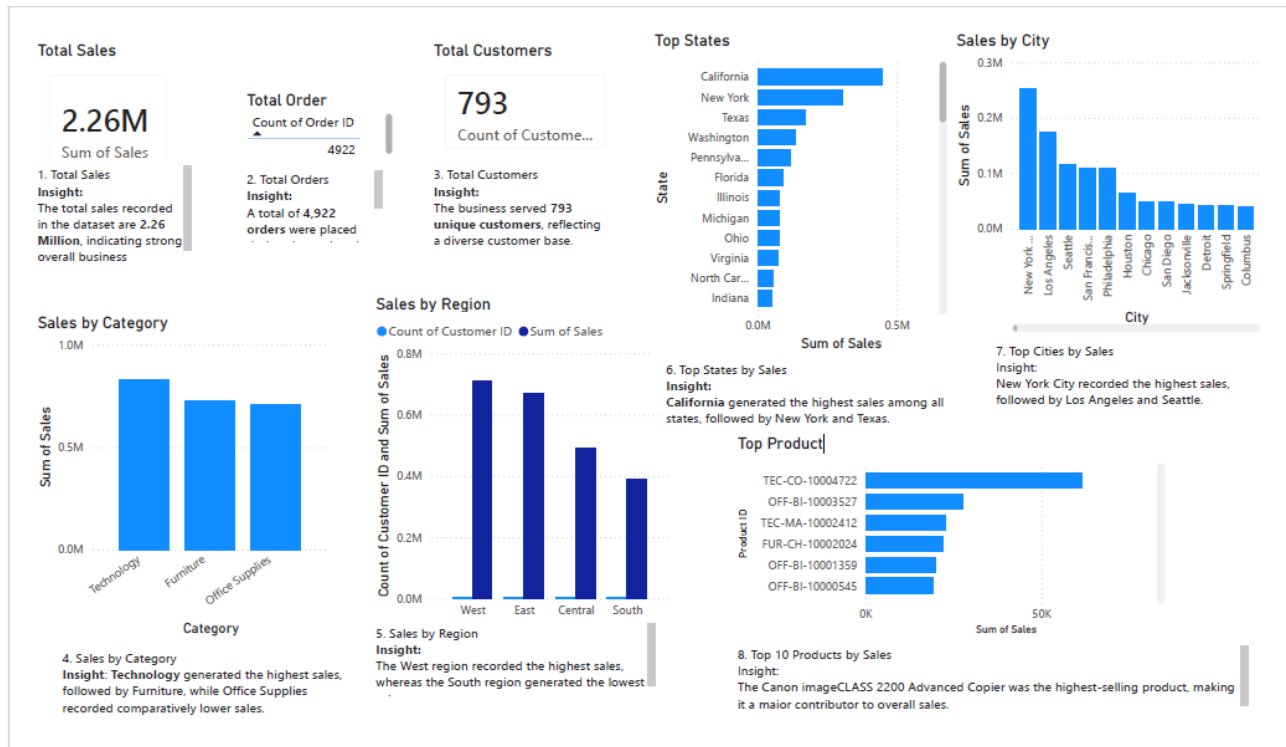


Figure 7.1: Executive Sales Performance Dashboard

The dashboard presents the overall sales performance using KPIs and visualizations.

It includes:

- Total Sales
- Total Orders
- Total Customers
- Sales by Category
- Sales by Region
- Top 10 States
- Top 10 Cities
- Top 10 Products

The dashboard enables users to identify high-performing products, regions, and customer trends.

8. SQL Analysis

Business analysis was performed using SQL queries in MySQL.

The following analyses were completed:

- Total Records :-

```
28
29 • SELECT COUNT(*) AS Total_Records
30 FROM clean_superstore;
31
```



The screenshot shows a SQL query result grid. The query is: `SELECT COUNT(*) AS Total_Records FROM clean_superstore;`. The result grid has a single row with the column name `Total_Records` and the value `9800`. The interface includes a 'Result Grid' tab, a 'Filter Rows' input field, and buttons for 'Export' and 'Wrap Cell Content'.

Total_Records
9800

Figure 8.1 Total Records

- Total Sales :-

```
11 • SELECT ROUND(SUM(Sales),2) AS Total_Sales
12 FROM clean_superstore;
13
```



The screenshot shows a SQL query result grid. The query is: `SELECT ROUND(SUM(Sales),2) AS Total_Sales FROM clean_superstore;`. The result grid has a single row with the column name `Total_Sales` and the value `2261536.78`. The interface includes a 'Result Grid' tab, a 'Filter Rows' input field, and buttons for 'Export' and 'Wrap Cell Content'.

Total_Sales
2261536.78

Figure 8.2 Total Sales

- Total Customers:-

```
14 • SELECT COUNT(DISTINCT `Customer ID`) AS Total_Customers
15 FROM clean_superstore;
16
```



The screenshot shows a SQL query result grid. The query is: `SELECT COUNT(DISTINCT `Customer ID`) AS Total_Customers FROM clean_superstore;`. The result grid has a single row with the column name `Total_Customers` and the value `793`. The interface includes a 'Result Grid' tab, a 'Filter Rows' input field, and buttons for 'Export' and 'Wrap Cell Content'.

Total_Customers
793

Figure 8.3 Total Customers

- Sales by Category:-

```

17 • SELECT Category,
18     ROUND(SUM(Sales),2) AS Total_Sales
19     FROM clean_superstore
20     GROUP BY Category
21     ORDER BY Total_Sales DESC;
22

```

Category	Total_Sales
Technology	827455.87
Furniture	728658.58
Office Supplies	705422.33

Figure 8.4 Sales by Category

- Sales by Region:-

```

23 • SELECT Region,
24     ROUND(SUM(Sales),2) AS Total_Sales
25     FROM clean_superstore
26     GROUP BY Region
27     ORDER BY Total_Sales DESC;
28

```

Region	Total_Sales
West	710219.68
East	669518.73
Central	492646.91
South	389151.46

Figure 8.5 Sales by Region

9. Key Insights

9.1 Total Sales

The total sales recorded are **2.26 Million**, indicating strong business performance.

9.2 Total Orders

The company processed **4,922 orders** during the analysis period.

9.3 Total Customers

The dataset contains **793 unique customers**.

9.4 Sales by Category

Technology generated the highest sales, followed by Furniture and Office Supplies.

9.5 Sales by Region

The West region generated the highest revenue, while the South region recorded the lowest sales.

9.6 Top States

California generated the highest sales, followed by New York and Texas.

9.7 Top Cities

New York City achieved the highest sales among all cities.

9.8 Top Products

Canon imageCLASS 2200 Advanced Copier generated the highest sales.

10. Business Recommendations

- Increase investment in Technology products.
- Improve marketing efforts in the South region.
- Maintain inventory for top-selling products.
- Monitor monthly sales trends.
- Use dashboard insights for strategic decision-making.

11. Conclusion

This project successfully demonstrates an end-to-end data analytics workflow using **Python, MySQL, and Power BI**.

Python was used for data cleaning and preprocessing, MySQL for business analysis and validation, and Power BI for interactive visualization. The dashboard provides valuable insights into sales performance across different business dimensions and supports data-driven decision-making.

12. Project Highlights

- End-to-End Data Analytics Workflow
- Python Data Cleaning & Preprocessing
- SQL Business Analysis using MySQL
- Interactive Dashboard using Power BI
- GitHub Portfolio Documentation

13. Future Scope

- Profit Analysis Dashboard
- Discount Analysis
- Customer Segmentation
- Sales Forecasting using Machine Learning
- Real-time Database Integration
- Interactive Drill-through Reports
- Mobile-friendly Dashboard

14. References

1. Kaggle – Superstore Sales Dataset
2. Microsoft Power BI Documentation
3. Pandas Documentation
4. NumPy Documentation
5. MySQL Documentation
6. Matplotlib Documentation

15. Key Skills Demonstrated

- Python Programming
- SQL
- MySQL
- Data Cleaning & Preprocessing
- Exploratory Data Analysis (EDA)
- Data Visualization
- Dashboard Development
- Power BI
- Business Intelligence
- Business Reporting